```
review for an interpolation of the movies <movie name 2> and Forrest
Gump (1994). Do not use the movies' names in your response.
```

where "<task name>" can be "summary", "positive review", "negative review", "neutral review", "five positive characteristics", "long description", and "pitch".

For the user profile task, we used zero-shot prompting like the following:

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In ten bullet points, generate an interpolation of the following
two user profiles:

User profile 1:

<user profile 1>

User profile 2:

<user profile 2>
```

## G Sample Outputs for Interpolations and CAV Extrapolations

Below we present sample results of interpolating "Forest Gump" with sample movies:

- Interpolation of "Inglorious Bastards (2009)" and "Forrest Gump (1994)" for the **five pos characteristics** task:
  - The movie is a gripping, suspenseful thriller about a man with a low IQ who achieves great things.
  - The movie is a powerful anti-war statement, showing the horrors of war and the toll it takes on those who fight in it.
  - The acting is superb, with all of the actors delivering powerful, memorable performances.
  - The cinematography is stunning, with the director creating some truly memorable visuals.
  - The story is well-written and engaging, with plenty of twists and turns to keep you guessing.
- Interpolation of "I, Robot (2004)" and "Forrest Gump (1994)" for the **long description** task:
  In 1983, a mentally challenged man named Charlie Baileygates is living in a small town in South Carolina. He is a kind and gentle soul, but he is also very naive. One day, Charlie meets a young woman named Myra who is new to town. Myra is kind to Charlie and they quickly become friends.

  One day, Charlie is walking through the woods when he comes across a crashed spaceship. Inside the spaceship, he finds a robot named Harry. Harry is damaged, but Charlie is able to fix him. Charlie and Harry quickly become friends, and Harry teaches Charlie about the world outside of their small town.

  One day, a government agent named Maiden comes to town looking for Harry. Maiden tells Charlie that Harry is a dangerous weapon, and that he must be destroyed. Charlie refuses to believe Maiden, and he and Harry run away.

  Maiden and her team of agents pursue Charlie and Harry across the country. Charlie and Harry are able to evade capture, but they are forced to go into hiding. Charlie and Harry eventually find a safe place to live, and they live happily ever after.

  The movie is a heartwarming story about friendship and acceptance. It is a reminder that everyone is capable of love and friendship, no matter what their circumstances. The movie is also a reminder that the world is a big and beautiful place, and that there is always something new to explore.

  The movie is well-acted and well-directed. The special effects are impressive, and the music is beautiful. The movie is a classic that will be enjoyed by generations to come.

> The movie is a reminder that we should never judge people by their appearance. Charlie may be mentally challenged, but he is also a kind and gentle soul. He is capable of love and friendship, and he deserves to be happy.
>
> The movie is also a reminder that we should never give up on our dreams. Charlie may be an unlikely hero, but he is able to overcome the odds and achieve his dreams.

We also demonstrate qualitative results of CAV extrapolations using the user profile task. In the following we want to move a user who doesn't really like funny movies to a viewer likes funny movies in the behavioral embedding space by following the CAV direction. Then we decode the extrapolated user embedding to get a user profile:

```
* They enjoy movies that are visually appealing and have a strong
  sense of style.
* They appreciate movies that are well-acted and have strong performances.
* They like movies that are thought-provoking and have a strong message.
* They enjoy movies that are suspenseful and have a lot of action.
* They like movies that are funny and have a lot of humor.
* They dislike movies that are too slow-paced or boring.
* They dislike movies that are too violent or gory.
* They dislike movies that are too dark or depressing.
* They dislike movies that are too predictable or formulaic.
```

The above user does *like movies that are funny and have a lot of humor* comparing with the groundtruth user profile which says "They are not a fan of light-hearted or comedic films.":

```
* They are a fan of foreign films.
* They enjoy films that are dark and suspenseful.
* They appreciate films with strong acting performances.
* They are interested in films that explore social and political issues.
* They are not a fan of light-hearted or comedic films.
* They do not enjoy films that are visually stunning but lack substance.
* They are not a fan of films that are set in the past.
* They do not enjoy films that are about war or violence.
* They are not a fan of films that are about crime or corruption."
```

## H Reinforcement Learning from AI Feedback

Reinforcement learning from AI feedback (RLAIF) is effective at aligning LMs to metrics that are labeled by off-the-shelf LMs in lieu of humans. Recent work like (Lee et al., 2023) has shown that utilizing a hybrid of human and AI preference models in conjunction with the *self-improving* fine-tuning technique outperforms the traditional supervised fine-tuned baselines and offers additional benefits from standalone RL fine-tunng with human feedback (RLHF). Utilizing RLAIF paradigms, one may fine-tune an ELM with a reward so that it would better align with consistency.

**Contextual Markov Decision Processes (CoMDPs)** The CoMDP is denoted by $(\mathcal{C}, \mathcal{S}, \mathcal{A}, P, r, s_0, N)$, in which the observable context space $\mathcal{C}$ contains both the user/item embedding vectors. The horizon $N$ is the length of the generated texts. For any $n < N$, the state space $\mathcal{S}$ at the $n$-th turn represents the sequence of tokens generated thus far $\{o_0, \ldots, o_{n-1}\}$, and the initial state $s_0$ is the initial start-of-sentence token $o_0$. The action space $\mathcal{A}$ is the language token vocabulary, with action $a \in \mathcal{A}$ representing any possible next token to be generated. The transition kernel $P$ models the next token distribution given the current sequence and contexts, which coincides with the LM policy, making the transition of our CoMDP known. Finally, the reward function $r$ measures the overall quality of the generated texts. The goal is to find a policy $\pi^*$ with maximum expected cumulative return, i.e., $\pi^* \in \arg\max_\pi J_\pi := \mathbb{E}[\sum_{n=0}^{N-1} r_t \,|\, P, s_0, \mathcal{C}, \pi]$. Note that the size of the tokenized state and action spaces grow exponentially with the vocabulary size.

Specifically, with text response $o = \{o_n\}_{n=0}^{N-1}$, and user-item embedding vectors $(w_u, w_v)$, the consistency reward is defined as follows:

$$r(o_n; o_{0:n-1}; w_v, w_u) = \begin{cases} \text{BC}(o; w_u)\ (\text{or SC}(o; w_v)) & \text{if } o_n = [\text{EOS}]; \\ 0 & \text{otherwise.} \end{cases}$$

The generation process of ELMs can be modeled using the following $N$-horizon CoMDP:

$$c = (w_v, w_u), \quad s_n = o_{0:n-1}, \quad a_n = o_n, \quad s_0 = o_0, \quad P(s_{n+1} \mid s_n, a_n) = \delta\{s_{n+1} = (s_n, a_n)\},$$

$$r(s_n, a_n; c) = \begin{cases} r(s_{n+1}; c) = r(o_{0:n}; w_v, w_u) & \text{if } n = N-1 \\ 0 & \text{otherwise} \end{cases}, \ \pi_\theta(a_n \mid s_n; c) = \mathbb{P}_\theta\big(o_n \mid o_{0:n-1}; w_v, w_u\big),$$

where $\delta_z$ denotes the Dirac distribution at $z$. Fine-tuning ELM is equivalent to maximizing the quality of the generated text given context, $\max_\theta\ \mathbb{E}_{(w_v,w_u)}\, \mathbb{E}_{\mathbb{P}_\theta(o_{0:N-1}|w_v,w_u)}[r(o; w_v, w_u)]$. The gradient of this objective function can be obtained as follows: $\nabla_\theta \mathbb{E}_{(w_v,w_u)}\, \mathbb{E}_{\mathbb{P}_\theta(o_{0:N-1}|w_v,w_u)}[r(o; w_v, w_u)] = \mathbb{E}_c\, \mathbb{E}_{\pi_\theta(\cdot|s_{0:N};c)} \left[ r(s_N; c) \sum_{n=0}^{N-1} \nabla_\theta \log \pi_\theta(s_n|a_n; c) \right]$. This is equivalent to applying the popular policy gradient algorithm REINFORCE to the aforementioned CoMDP for personalized text generation. The gradient of the objective function is estimated using trajectories $\prod_{n=0}^{N-1} \pi_\theta(s_n|a_n; c)$ generated by the current policy, and then used to update the ELM policy in an online fashion.

**Adding KL regularization:** We add the KL between the fine-tuned and pre-trained models as a regularizer to the objective function. Leveraging the auto-regressive nature of ELMs one can compute the KL regularization over the entire sequence/trajectory (of tokens), i.e., $\text{KL}\big(\mathbb{P}_\theta(o_{0:N-1}|w_v, w_u) \| \mathbb{P}_{\text{pre}}(o_{0:N-1}|w_v, w_u)\big)$. The resulting objective function is as follows:

$$\max_\theta\ J(\theta) := \mathbb{E}_{(w_v,w_u)}\, \mathbb{E}_{\mathbb{P}_\theta(o_{0:N-1}|w_v,w_u)} \left[ r(o_{0:N-1}; w_v, w_u) - \beta \log \frac{\mathbb{P}_\theta(o_{0:N-1}|w_v, w_u)}{\mathbb{P}_{\text{pre}}(o_{0:N-1}|w_v, w_u)} \right]. \tag{1}$$

It can be shown that this problem is equivalent to the KL-regularized objective in the CoMDP.

Denote by $\mathcal{D}$ a replay buffer of trajectories $\{(w_v, w_u, o_{0:N-1})\}$ generated by arbitrary ELMs $\mathbb{P}_{\theta'}(o_{0:N-1}|w_v, w_u)$ and $\tau = \{(c, s_n, a_n, s_{n+1})\}_{n=0}^{N-1} \sim \mathcal{D}$ a trajectory sampled from the offline data $\mathcal{D}$, where $(s_n, a_n, s_{n+1})$ is a tuple of state, action, and next state of the CoMDP, respectively. With this KL regularization we can utilize the *soft actor critic* framework (Haarnoja et al., 2018) to develop RL updates for the *value function* $\{V_n(s; c)\}_{n=0}^{N-1}$, *state-action value function* $\{Q_n(s, a; c)\}_{n=0}^{N-1}$, and *ELM policy* $\prod_{n=0}^{N-1} \pi_\theta(s_n|a_n; c)$ (initialized with $\prod_{n=0}^{N-1} p_{\text{pre}}(s_n|a_n; c)$) that minimize losses:

$$L_Q = \mathbb{E}_{\tau\sim\mathcal{D}} \left[ \sum_{n=0}^{N-2} (V_{\text{tar},n+1}(s_{n+1}; c) - Q_n(s_n, a_n; c))^2 + (r(s_N; c) - Q_{N-1}(s_{N-1}, a_{N-1}; c))^2 \right], \tag{2}$$

$$L_V = \mathbb{E}_{\tau\sim\mathcal{D}} \left[ \sum_{n=0}^{N-1} (Q_{\text{tar},n}(s_n, a_n; c) - \alpha \log \frac{\pi_\theta(a_n|s_n; c)}{p_{\text{pre}}(a_n|s_n; c)} - V_n(s_n; c))^2 \right], \tag{3}$$

$$L_\theta = \mathbb{E}_{\tau\sim\mathcal{D}} \left[ \sum_{n=0}^{N-1} Q_n(s_n, a_n; c) - \alpha \log \frac{\pi_\theta(a_n|s_n; c)}{p_{\text{pre}}(a_n|s_n; c)} \right], \tag{4}$$

where the critic $Q_n$ and $V_n$ take any token sequences at step $n$ as input and predict the corresponding cumulative return; $\alpha > 0$ is the entropy temperature; $(V_{\text{tar},n}, Q_{\text{tar},n})$ are the target value networks.

Besides iteratively updating the ELM policies and their critic functions, consider the closed-form optimal solution of the Bellman equation of this entropy-regularized RL problem:

$$V_n^*(s; c) = \alpha \cdot \log \mathbb{E}_{a\sim p_{\text{pre}}(\cdot|s;c)}[\exp(\frac{Q_n^*(s, a; c)}{\alpha})],\ \forall n, \tag{5}$$

$$Q_{N-1}^*(s, a; c) = r(s; c),\ Q_n^*(s, a; c) = \mathbb{E}_{s'\sim P(\cdot|s,a)}[V_{n+1}^*(s'; c)],\ \forall n < N-1, \tag{6}$$

$$\mu_n^*(a|s; c) = p_{\text{pre}}(a|s; c) \cdot \exp(\frac{Q_n^*(s, a; c)}{\alpha})\ /\ \mathbb{E}_{a\sim p_{\text{pre}}(\cdot|s;c)}[\exp(\frac{Q_n^*(s, a; c)}{\alpha})],\ \forall n, \tag{7}$$

where the time-dependent optimal policy (at time $n$), i.e., $\mu_n^*$ is a softmax policy w.r.t. the optimal state-action values $Q_n^*$ over different actions sampled from the pre-trained ELM $p_{\text{pre}}$. Therefore,